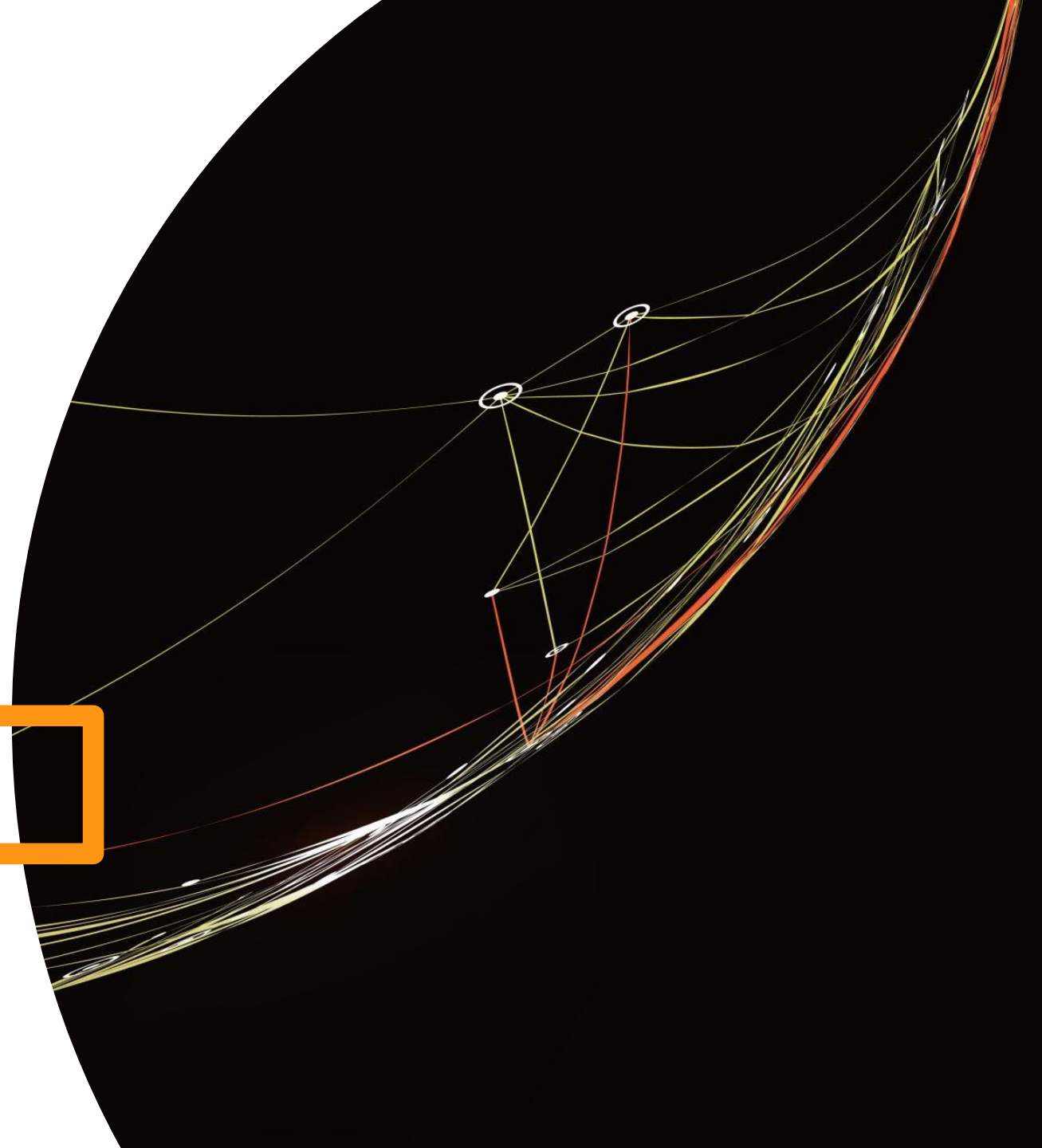


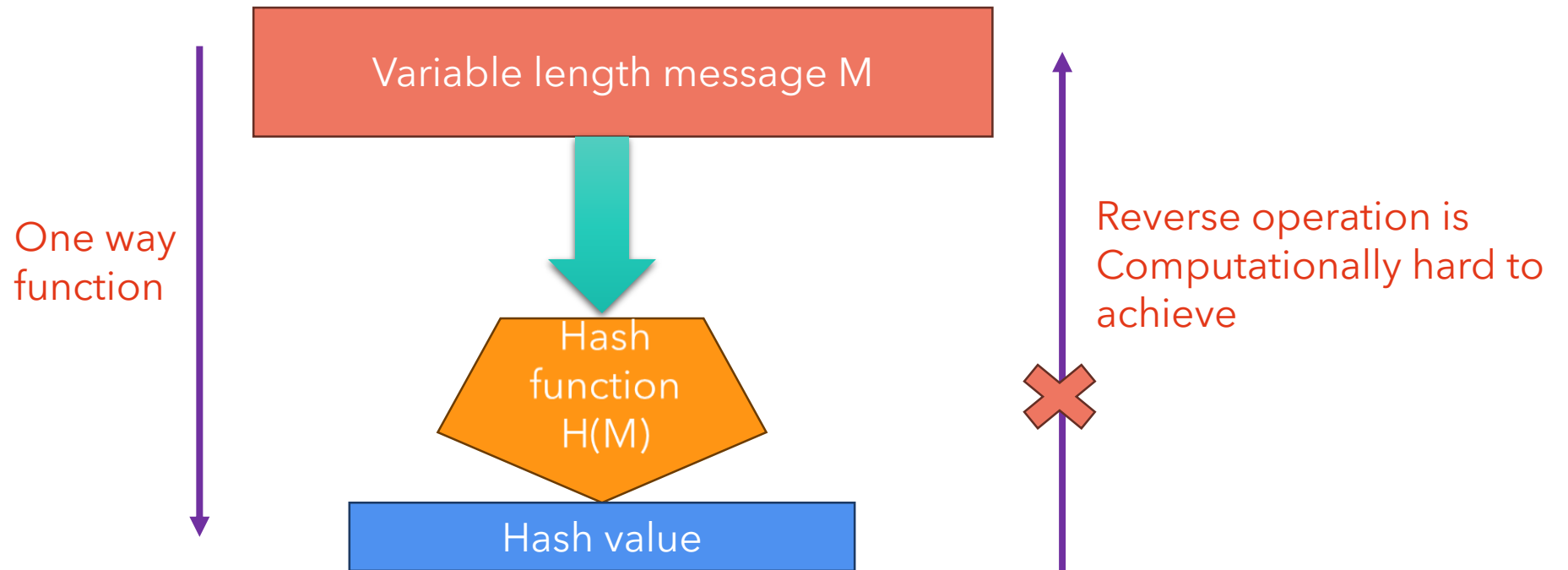
Introduction to hash functions

Module IV



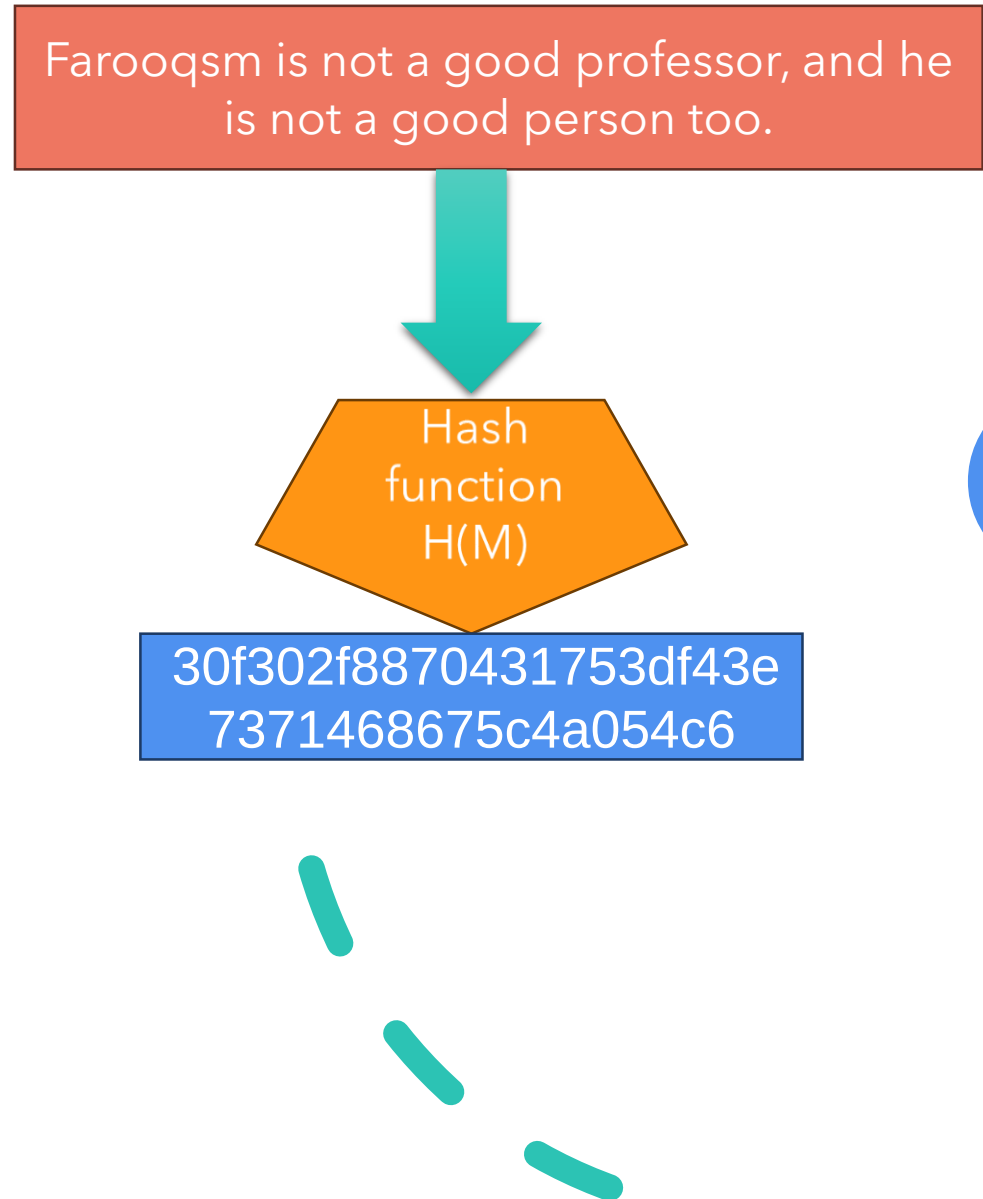
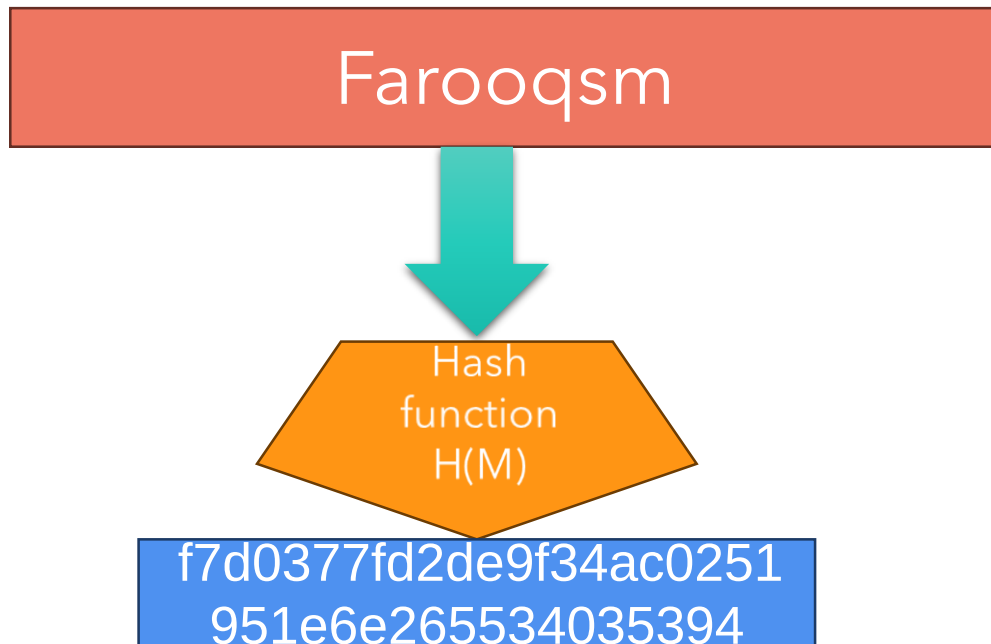
Hash function

- A hash function H accepts a variable length block of data M as Input and produces a fixed size hash value.



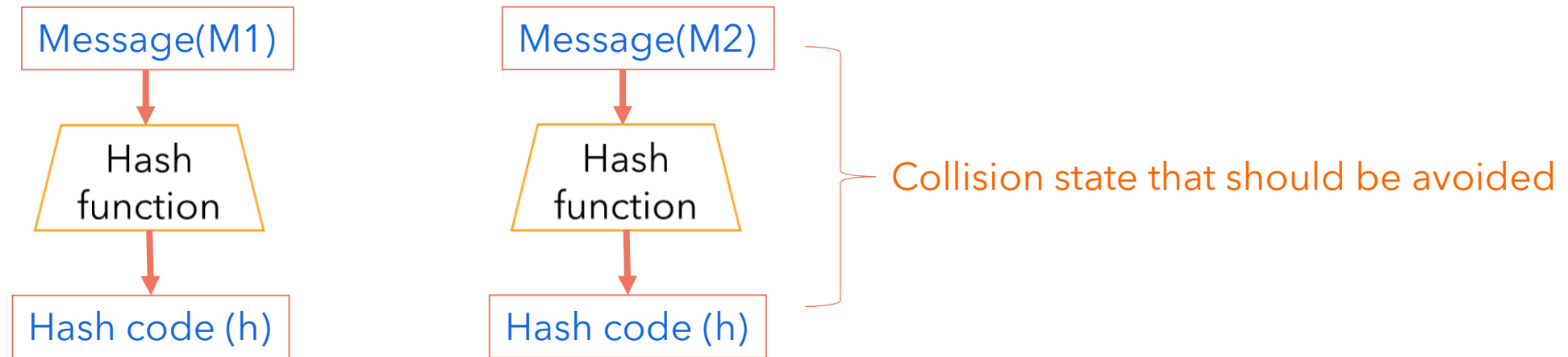
Hash function example

- Variable size Inputs but fixed output.



Properties of a hash function

- A good hash function has a property that the results of applying the function to a large set of inputs will produce outputs that are evenly distributed and apparently random.
- A change to any bit or bits in results, with high probability, in a change to the hash code. (**Avalanche effect**)
- Hash function should be a **collision resistant**.



Cryptographic hash functions

- Hash functions used for security applications are known as cryptographic hash functions.
- Two Important properties:-
- It is computationally infeasible to find either,
 - a) a data object that maps to a pre-specified hash result
(The one-way property) or
 - b) Two data objects that map to the same hash result.
(The collision free property)

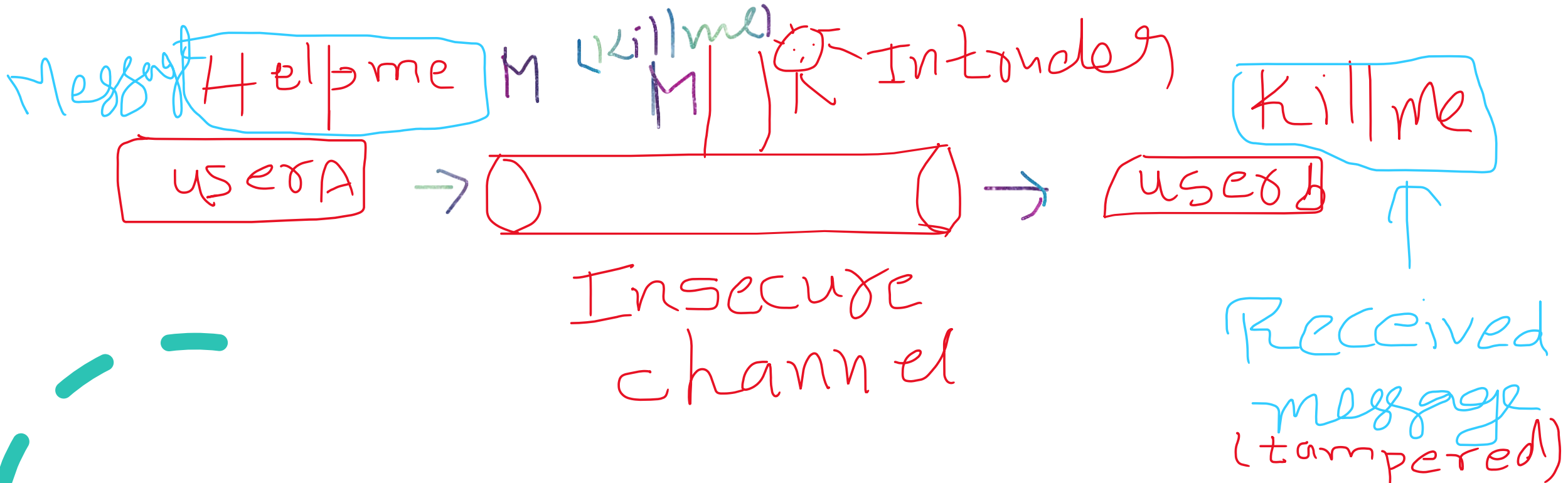
[Intruder try to detect collisions only through brute-force method.]

Applications of hash function in cryptography

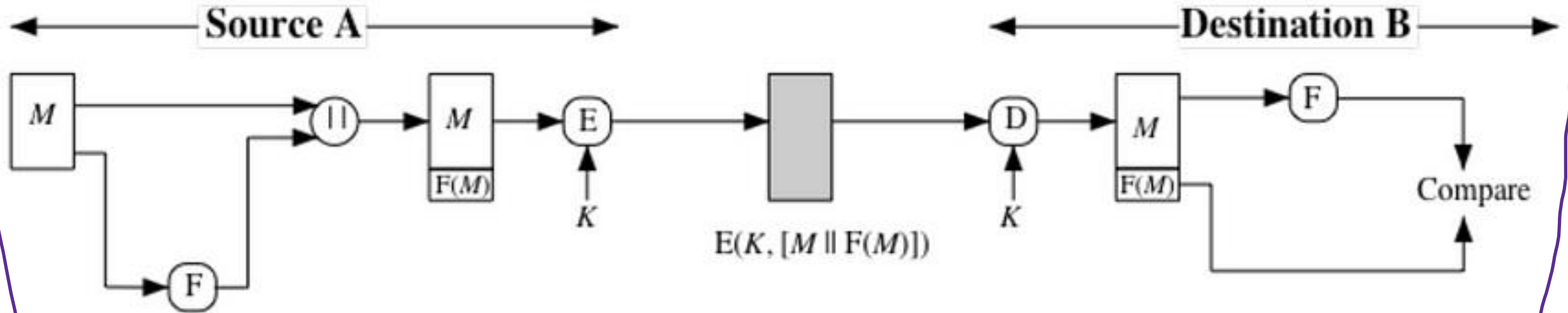
- Message authentication
- Digital signature
- One way password
- Intrusion detection and prevention

Message authentication

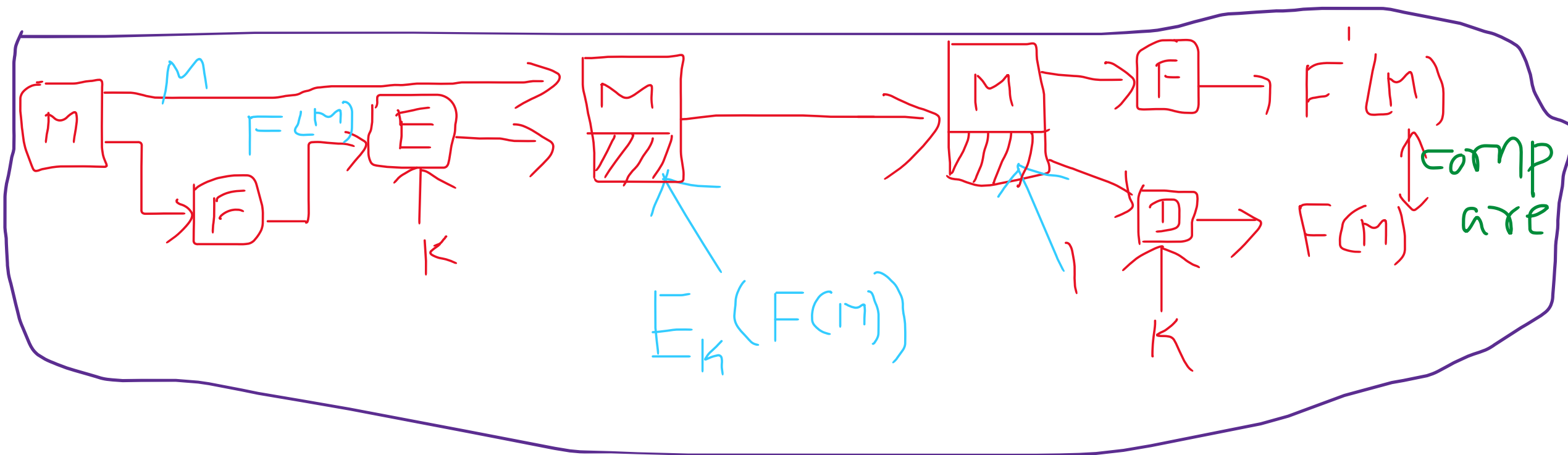
- It is a mechanism or service used to verify the Integrity of a message.
- Message authentication assures data received are exactly as sent (i.e., contain no modification, Insertion, deletion or replay).



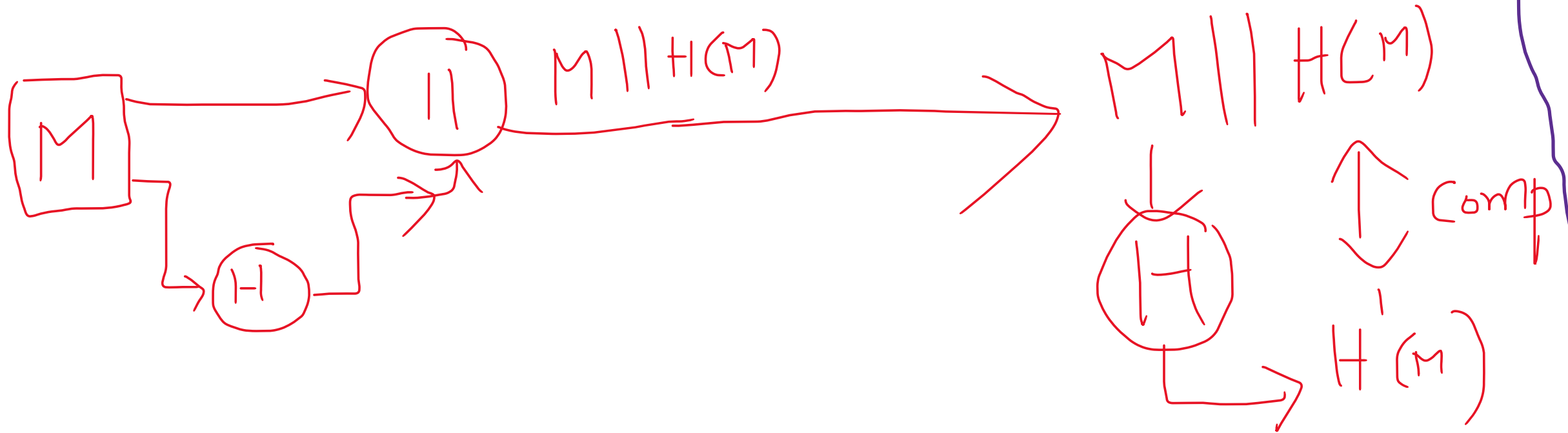
Method-I (Message and hash code encrypted)



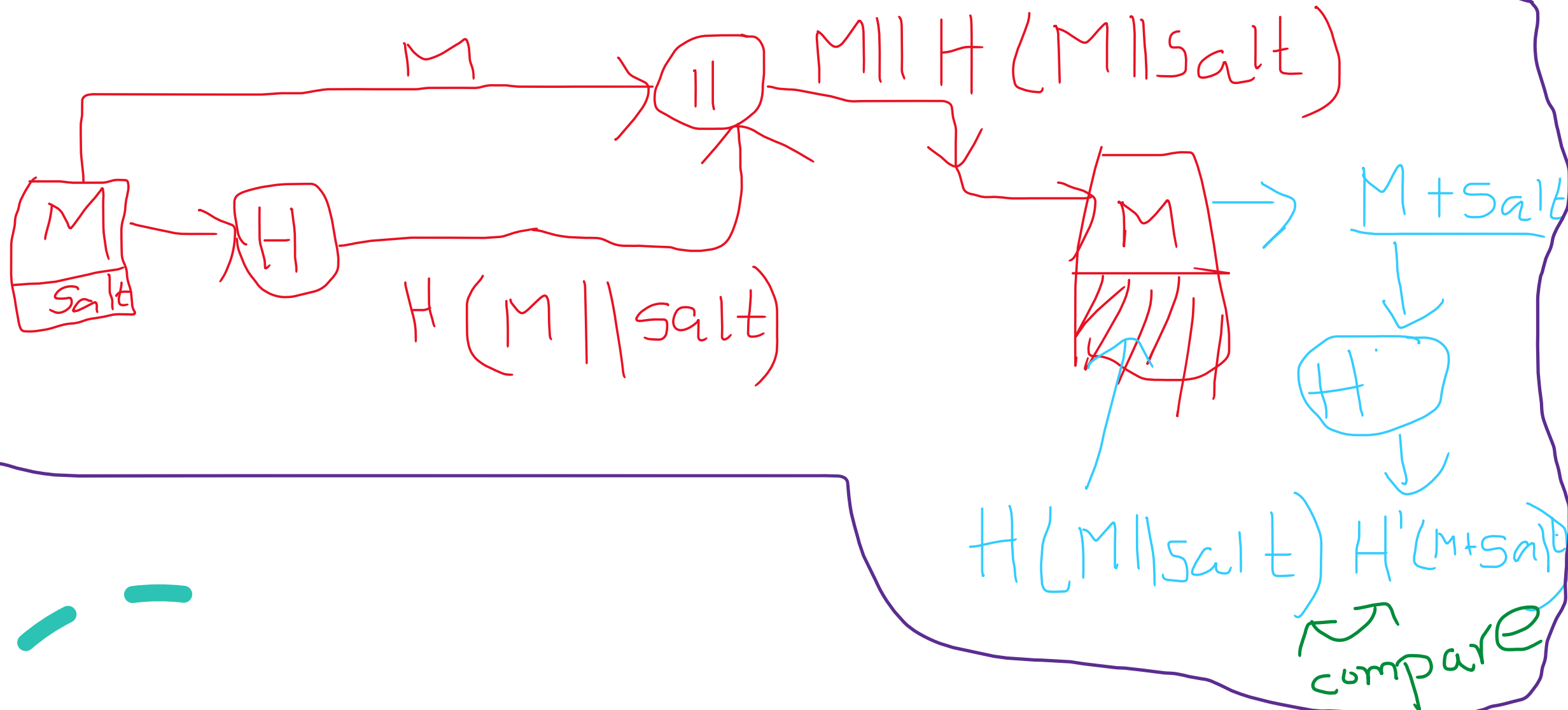
Method-II



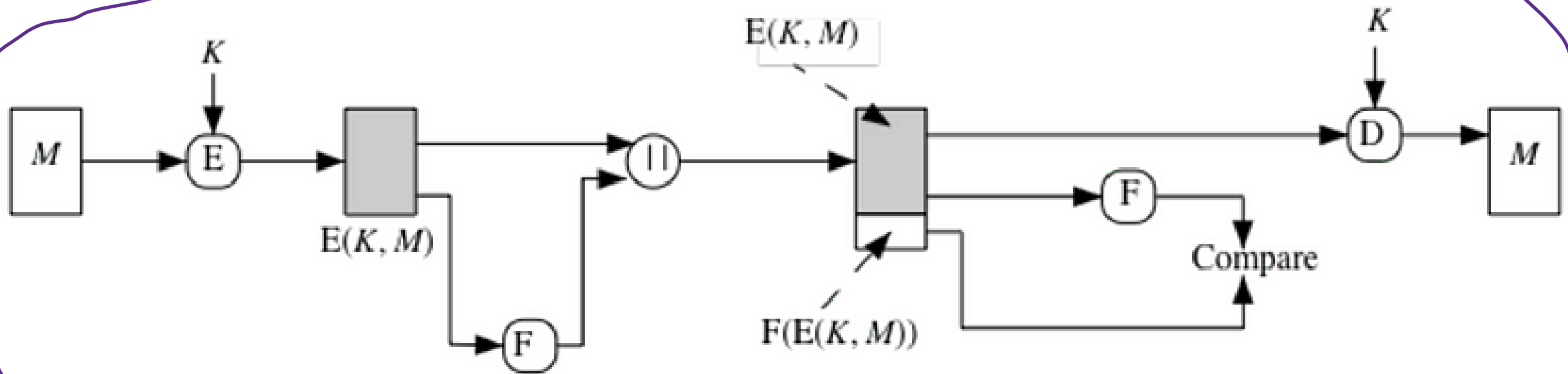
Method-III



Method-IV



Method-V



Security requirements for cryptographic hash functions

Requirement	Description
Variable size	Hash function (H) can be applied to a block of data of any size.
Fixed output size	Hash function (H) produces a fixed length output.
Efficiency	$H(x)$ is relatively easy to compute for any given x , making both hardware and software implementations practical's.
Pre-Image resistant	For any given hash value h , it is computationally infeasible to find y such that $H(y) = h$.
Second pre-Image resistant (weak collision resistant)	For any given block x , it is computationally infeasible to find $y \neq x$ with $H(y) = H(x)$.
Collision resistant (Strong collision resistant)	It is computationally infeasible to find any pair (x, y) such that $H(x) = H(y)$.
Pseudorandomness	Output of Hash function (H) meets standard tests for pseudorandomness.

Pre-Image resistant

- $H(\text{Ahmad}) = \text{ABC20DF}$

↓
Hacker knows
hash value
but couldn't
get its pre-Image
→ Ahmad
(PreImage)

Second pre-Image resistant

$H(\text{Ahmad}) = \text{ABC20DF}$



Hacker

knows hashvalue

$H(\text{Farooq})$



ABC20DF

but he couldn't find
another pre-Image which gives the same
hash value